Neural Network Classification of Corneal Topography

Preliminary Demonstration

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Purpose. Videokeratography is a powerful tool for the diagnosis of corneal shape abnormalities. However, interpretation of the topographic map is sometimes difficult, especially when pathologies with similar topographic patterns are suspected. The neural networks model, an artificial intelligence approach, was applied for automated pattern interpretation in corneal topography, and its usefulness was assessed.

Methods. One hundred eighty-three topographic maps were selected and classified by human experts into seven categories: normal, with-the-rule astigmatism, keratoconus (mild, moderate, advanced), postphotorefractive keratectomy, and postkeratoplasty. The maps were divided into a training set (108 maps) and a test set (75 maps). For each map, 11 topography-characterizing indices calculated from the data provided by the TMS-1 videokeratoscope, plus the corresponding diagnosis category, were used to train a neural network.

Results. The correct classification was achieved by a trained neural network for all 108 maps in the training set. In the test set, the neural network correctly classified 60 of 75 maps (80%). For every category, accuracy and specificity were greater than 90%, whereas sensitivity ranged from 44% to 100%.

Conclusions. With further testing and refinement, the neural networks paradigm for computer-assisted interpretation or objective classification of videokeratography may become a useful tool to aid the clinician in the diagnosis of corneal topographic abnormalities. Invest Ophthalmol Vis Sci. 1995;36:1327–1335.

Keratometry and photokeratoscopy are useful to evaluate topographic abnormalities of the cornea. However, it is difficult to assess the precise characteristics of corneal shape by means of keratometry readings or inspection of mire images alone. Computer-assisted data analysis[1] and graphic presentation in the form of color-coded maps[2] have greatly expanded the information obtainable from videokeratography. Consequently, this procedure has become the only clinically available examination that can detect subtle topographic alterations of the cornea, such as early keratoconus,[3] contact lens-induced corneal warpage,[4] diurnal fluctuation of topography after radial keratotomy,[5] and decentration of the transparent optical zone after photorefractive keratectomy.[6]

Although videokeratography is considered the most reliable means of assessing corneal topography, visual interpretation of color-coded maps is essentially subjective. Acquiring skill in the interpretation of topographic maps requires an understanding of the principles and techniques of videokeratography as well as broad exposure to a variety of clinical entities. Because the diversity of topographic patterns can make interpreting maps difficult, automated or objective systems might be expected to aid interpretation of corneal topography.

Recently, artificial intelligence has been applied to medical diagnostic instruments as a computer-assisted decision support system. In the electrocardiogram, for example, the best of the commercially available interpretation systems can outperform most phy-

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TABLE 1. Criteria for Topographic Pattern Categorization

Normal Cornea
Clinically diagnosed normal eye
Topographic enantiomorphism in both eyes
Radial symmetry and peripheral flattening in all semimeridians
No abnormal flattening or steepening
Absence of irregular astigmatism
Keratometric cylinder < 1.5 D
Astigmatism With the Rule
Clinically diagnosed with-the-rule astigmatism
Keratometric cylinder ≥ 1.5 D
Symmetrical, vertically aligned bowtie pattern
Peripheral flattening in all semimeridians
No abnormal flattening or steepening
Absence of irregular astigmatism
Keratoconus (subjectively divided into mild, moderate, and severe)
Clinically diagnosed keratoconus
Abnormal localized area of steepening
Postphotorefractive keratectomy
Postphotorefractive keratectomy eye
Well-centered ablation area
Peripheral steepening inside the ablation area and peripheral flattening outside the ablation area
Postkeratoplasty
Postpenetrating keratoplasty eye

sicians and may perform nearly as well as expert cardiologists.[7] Neural network models consist of hardware or software that simulates some common aspects of the biologic nervous system. This artificial intelligence approach was originally used by neuroscientists to model human cognition. Because neural networks can interpolate or predict complicated data through "learning" by simulating human neurologic processing abilities, they have been attracting considerable attention in science, economics, and industry. Neural networks have been applied in medicine not only to classify electrocardiographic findings[7] but also to diagnose liver diseases or malignancies from laboratory data,[8,9] to estimate fetal weight by ultrasonography,[10] and to interpret visual fields.[11,12]

Classification of corneal topography is one area of ophthalmology in which neural networks could be used. This is, to our knowledge, the first report of the successful application of a neural network model to classify corneal topography.

METHODS

Subjects

Corneal topographic maps were obtained from the patient population at the Louisiana State University Eye Center with a computer-assisted videokeratoscope (TMS-1, Computed Anatomy, New York, NY). Institutional review board approval was obtained, all patients signed informed consent forms, and the tenets of the Declaration of Helsinki were followed in taking videokeratographs.

The 1.5-diopter interval scale (Klyce–Wilson scale) was used to display the color-coded maps.[13] The three authors classified typical maps into seven categories based on the topographic appearance of the maps and clinical records (Table 1). The categories were normal; with-the-rule astigmatism; mild, moderate, and advanced keratoconus; postphotorefractive keratectomy, and postkeratoplasty. Maps in which focusing or alignment were not properly achieved or that contained atypical topographic appearances were excluded from the study. For this project, 183 eyes were selected. The maps were randomly divided into a training set (108 maps) and a test set (75 maps), as shown in Table 2.

Topographic Indices

Eleven quantitative descriptors from each map were used as input data for the neural network model. These indices were selected and devised to extract the characteristics of corneal topographic patterns. Each of the indices was found subsequently to have significant weight in the neural network. All indices, except the standard deviation of power (SDP), have been published.[14–16] Briefly, the Simulated Keratometry value (SimK) is determined from the power on mires 7, 8, and 9 of the videokeratoscope for 128 equally spaced meridians.[14] SimK1 is the maximum power value defining the major axis, and SimK2 represents the average power value associated with the meridian orthogonal to SimK1. MinK is defined as the minimum average power of rings 7, 8, and 9. The Surface Asymmetry Index (SAI) is defined as the centrally weighted summation of corneal power differences between corresponding points 180° apart on the mires, over 128 equally spaced meridians.[15] The Surface Regularity Index (SRI) is a measure of local fluctuations in corneal power. It is centrally weighted and calculated with the data from the inner 10 rings.[14] For the Differential Sector Index (DSI) and the Opposite Sector Index (OSI),[16] the measured cor-

TABLE 2. Subjects in Each Set

	Training Set	*Test Set*	*Total*
Normal cornea	18	12	30
With-the-rule astigmatism	17	11	28
Keratoconus (mild)	12	8	20
Keratoconus (moderate)	14	12	26
Keratoconus (advanced)	13	9	22
Postphotorefractive keratectomy	17	11	28
Postkeratoplasty	17	12	29
Total	108	75	183

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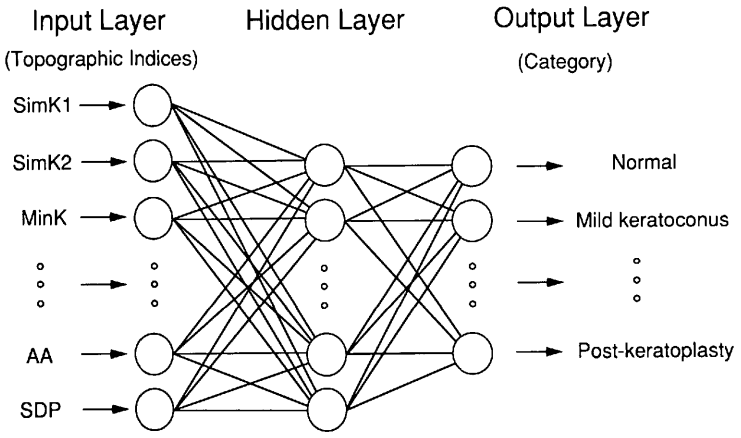

FIGURE 1. Schematic diagram of neural network architecture. This model consists of three layers of neurons that are fully connected. Neurons in the input layer and the output layer correspond to the topographic indices and the topographic categories, respectively. Although not all are shown in this diagram, there are 11 neurons in the input layer, 18 neurons in the hidden layer, and 7 neurons in the output layer.

neal surface is divided into eight pie-shaped sectors, each subtending an angle of 45°. The average power is calculated for each sector based on the area-corrected corneal power. New sector regions are defined by rotating the pattern in 1.41° steps, up to a maximum of 45°, and recalculating the average power in each sector at each step. The DSI reports the greatest difference in the average powers between any two sectors. OSI represents the maximum difference between average powers in opposite sectors. The Center–Surround Index (CSI) is the difference in the average area-corrected corneal power between the central area (3.0 mm diameter) and an annulus surrounding the central area (3.0 to 6.0 mm).[16] The Irregular Astigmatism Index (IAI) is an average summation of inter-ring power variations along every semimeridian for the entire analyzed corneal surface.[16] The Analyzed Area (AA) is the ratio of the interpolated data area to the total analyzed area.[16] The new index, Standard Deviation of Power (SDP), is the standard deviation of the corneal power array with area correction.

Neural Network Model

The backpropagation method[17] was used as a neural network model. The backpropagation model consists

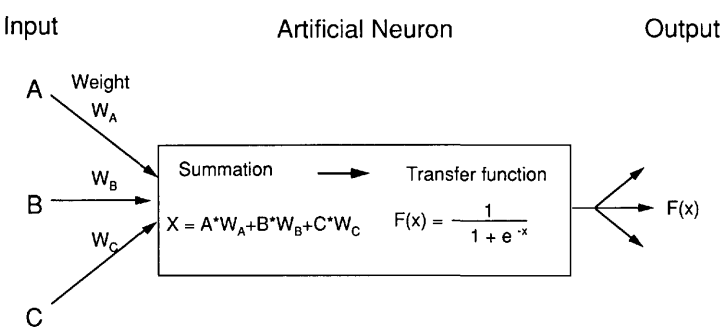

FIGURE 2. Schematic diagram of a neural network neuron. Each of the input values (**A, B, C**) is multiplied by a weight (W_A, W_B, W_C, respectively). The sum of weighted inputs is transformed by a nonlinear transfer function, and the resultant output is used as the input for the next neuron.

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TABLE 3. Average and Standard Deviation of All Indices

	NRM (n = 30)	*AST* (n = 28)	*KC1* (n = 20)	*KC2* (n = 26)	*KC3* (n = 19)	*PRK* (n = 28)	*KP* (n = 29)
SimK1	43.94 ± 1.02	45.33 ± 1.62	48.12 ± 2.37	54.27 ± 3.76	56.34 ± 6.95	42.22 ± 1.43	49.84 ± 3.99
SimK2	43.32 ± 1.08	42.40 ± 1.29	44.97 ± 1.75	47.94 ± 3.12	47.93 ± 4.94	41.54 ± 1.33	41.73 ± 3.13
MinK	43.27 ± 1.07	42.37 ± 1.28	44.68 ± 1.76	47.68 ± 3.16	45.37 ± 4.40	41.46 ± 1.35	41.16 ± 3.21
SAI	0.26 ± 0.08	0.31 ± 0.12	1.45 ± 0.63	2.14 ± 0.72	5.11 ± 2.09	0.29 ± 0.07	1.68 ± 0.96
SRI	0.28 ± 0.14	0.55 ± 0.26	1.01 ± 0.49	1.42 ± 1.39	2.99 ± 1.53	0.43 ± 0.17	1.58 ± 0.53
DSI	1.86 ± 0.55	3.84 ± 1.34	6.09 ± 2.10	11.74 ± 2.49	12.96 ± 6.85	1.43 ± 0.47	9.26 ± 3.37
OSI	0.77 ± 0.44	0.58 ± 0.45	4.78 ± 2.09	9.91 ± 2.90	8.15 ± 5.92	0.58 ± 0.38	3.70 ± 1.77
CSI	0.26 ± 0.18	0.23 ± 0.22	1.16 ± 1.09	2.67 ± 1.83	−0.35 ± 3.19	−0.99 ± 0.42	−0.45 ± 1.70
IAI	0.34 ± 0.05	0.39 ± 0.09	0.42 ± 0.12	0.50 ± 0.08	0.83 ± 0.17	0.37 ± 0.06	0.68 ± 0.15
AA	81.61 ± 4.06	78.53 ± 6.47	80.00 ± 5.99	80.58 ± 7.97	48.49 ± 13.40	81.96 ± 4.21	46.27 ± 12.25
SDP	0.84 ± 0.16	1.42 ± 0.45	2.65 ± 0.85	4.94 ± 1.07	6.25 ± 2.38	0.95 ± 0.30	3.69 ± 1.18

NRM = normal cornea; AST = with-the-rule astigmatism; KC1 = mild keratoconus; KC2 = moderate keratoconus; KC3 = advanced keratoconus; PRK = postphotorefractive keratectomy; KP = postkeratoplasty.

of three or more layers of the basic processing unit, called a "neuron." Each neuron in a preceding layer is connected with every neuron in the following layer (Fig. 1). The input layer consists of investigator-defined neurons analogous to sensory inputs. The output layer is made up of neurons that correspond to classification categories. Between the input layer and the output layer are one or more interconnected hidden layers.

Each neuron beyond the input layer receives a series of weighted inputs from neurons of the previous layer. The sum of the weighted inputs to a given neuron is transformed by a nonlinear transfer function (sigmoidal curve) to modulate the response characteristics (Fig. 2).

In the training process, the neural network compares its output response with the correct response and produces an error value. The error value is then used to adjust the weighting scheme to minimize the output error. Each pattern of inputs from a map and its corresponding desired response are applied to the network iteratively until the error of the whole training set reaches a predefined threshold tolerance value. When a neural network completes its training, the relationship between input and output data is memorized as a matrix of adjusted weights. Once a neural network is well trained, it can be applied to unknown data to make rapid predictions.

Architectural Parameters and Training Method of the Neural Network

The network was constructed with three layers (Fig. 1). The input layer consisted of 11 neurons, each of which was assigned to one of the topographic indices (SimK1, SimK2, MinK, SAI, SRI, DSI, OSI, CSI, IAI, AA, SDP). There was one hidden layer of 18 neurons. The output layer consisted of seven neurons, one for each topographic category (normal; with-the-rule astigmatism; mild, moderate, and advanced keratoconus; postphotorefractive keratectomy; and postkeratoplasty). The nonlinear transfer functions for the neurons in the hidden and output layers were normalized to provide an output ranging from 0 to 1. As a result, a value of 0 for a particular output neuron

TABLE 4. Results of a Neural Network Classification of the Corneal Topography Patterns in the Training Set

	Output of Neural Network							
Actual Category	*NRM*	*AST*	*KC1*	*KC2*	*KC3*	*PRK*	*KP*	*Unknown*
NRM	18	0	0	0	0	0	0	0
AST	0	17	0	0	0	0	0	0
KC1	0	0	12	0	0	0	0	0
KC2	0	0	0	14	0	0	0	0
KC3	0	0	0	0	13	0	0	0
PRK	0	0	0	0	0	17	0	0
KP	0	0	0	0	0	0	17	0

NRM = normal cornea; AST = with-the-rule astigmatism; KC1 = mild keratoconus; KC2 = moderate keratoconus; KC3 = advanced keratoconus; PRK = postphotorefractive keratectomy; KP = postkeratoplasty.

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TABLE 5. Results of Neural Network Classification of Corneal Topography Patterns in the Test Set

	Output of Neural Network							
Actual Category	*NRM*	*AST*	*KC1*	*KC2*	*KC3*	*PRK*	*KP*	*Unknown*
NRM	10	0	0	0	0	0	0	2
AST	1	9	1	0	0	0	0	0
KC1	0	2	5	0	0	0	0	1
KC2	0	0	1	11	0	0	0	0
KC3	0	0	0	1	4	0	3	1
PRK	0	0	0	0	0	11	0	0
KP	0	0	0	0	1	0	10	1

NRM = normal cornea; AST = with-the-rule astigmatism; KC1 = mild keratoconus; KC2 = moderate keratoconus; KC3 = advanced keratoconus; PRK = postphotorefractive keratectomy; KP = postkeratoplasty.

indicated a complete lack of response to the corresponding category, whereas an output strength approaching 1 indicated maximum response. The parameters that were used in this network were set initially from the values calculated from our input–output architecture according to the software developer's strategy (learning rate, 100%; smoothing factor, 0.9; number of hidden layers, 1). The number of hidden neurons was estimated and varied to optimize the results within the constraints of the overall architecture.

The learning process of the neural network was performed using the data set of 11 topographic indices and the corresponding seven topographic categories in the training set. The maps of the training set were randomly ordered before they were distributed to the network. A single training run was defined as the presentation of the data in all 108 maps. Training of the neural network was considered complete when the values of the correct output neurons were between 0.9 and 1.0 and the values for the other (incorrect) categories were between 0 to 0.1 for all training maps (training tolerance, 0.1).

The neural network model developed in our study was implemented on a 80486-based personal computer using the Brainmaker Professional software package (version 2.53; California Scientific Software, Nevada City, CA).

Evaluation of Neural Network Efficacy With the Test Set

The efficacy of the trained neural network was evaluated with the test set: 75 maps, not including any that were used for training. In this part of the study, an output strength above the cut-off value (0.5) of a particular output neuron was defined as a positive response of the neural network to the corresponding category, and an output strength less than 0.5 was defined as a lack of response to the category.[7,9] If multiple output neurons were above 0.5, the category that corresponded to the neuron with the greatest output was considered to be the interpretation of the map. When no output neuron had a value > 0.5 for a particular map, the network was assumed to give no interpretation for that map (i.e., classified as unknown pattern).[7] The results were described for each category in terms of sensitivity (true positive/[true positive + false negative]), specificity (true negative/[true

TABLE 6. Efficacy of the Neural Network in the Test Set

Category	(n)	*Sensitivity (%)*	*Specificity (%)*	*Accuracy (%)*
NRM	12	10/12 (83)	62/63 (98)	72/75 (96)
AST	11	9/11 (82)	62/64 (97)	71/75 (95)
KC1	8	5/8 (63)	65/67 (97)	70/75 (93)
KC2	12	11/12 (92)	62/63 (98)	73/75 (97)
KC3	9	4/9 (44)	65/66 (98)	69/75 (92)
PRK	11	11/11 (100)	64/64 (100)	75/75 (100)
KP	12	10/12 (83)	60/63 (95)	70/75 (93)
Total	75			60/75 (80)

NRM = normal cornea; AST = with-the-rule astigmatism; KC1 = mild keratoconus; KC2 = moderate keratoconus; KC3 = advanced keratoconus; PRK = postphotorefractive keratectomy; KP = postkeratoplasty.

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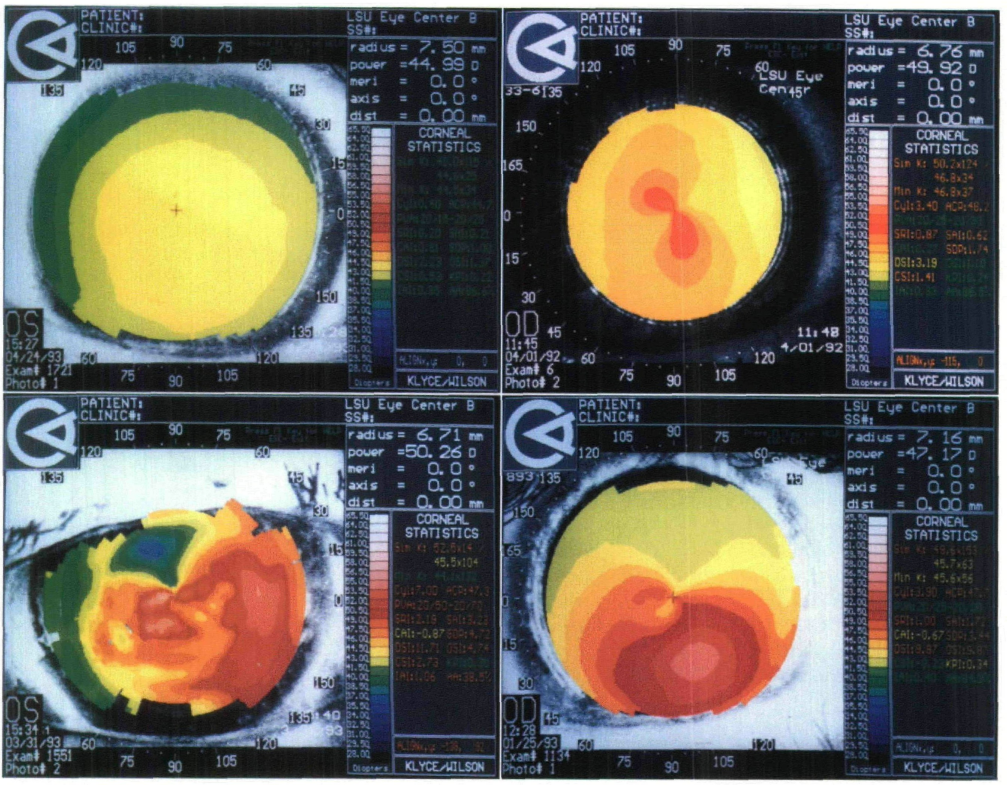

negative + false positive]), and accuracy ([true positive + true negative]/total number of maps).

RESULTS

Basic statistics of the topographic indices that were used as input data for the neural network are shown for all categories in Table 3. Each category showed a unique distribution of indices, which enabled discrimination by the neural network.

The training of the neural network was completed in 1509 iterations of runs with a training tolerance of 0.1. The neural network correctly classified all 108 maps in the training set (Table 4). The correspondence between the actual categories and the classifications obtained with the neural network for the test set is shown in Table 5. The network correctly classified 60 of the 75 maps (overall accuracy, 80%). Sensitivity, specificity, and accuracy in each category are shown in Table 6. Although specificity and accuracy for every category were greater than 90%, sensitivity was variable (advanced keratoconus 44%, mild keratoconus 63%, astigmatism 82%, postkeratoplasty 83%, normal 83%, moderate keratoconus 92%, and post-photorefractive keratectomy 100%).

Examples of color-coded maps in the test set and their corresponding output strength of the neural network are shown in Figures 3 through 6. Fifteen maps in the test set were not correctly classified (Table 5). Two maps of keratoconus were misclassified with respect to severity. Two maps of mild keratoconus were classified as astigmatism, and one astigmatism map was classified as mild keratoconus. Three maps of advanced keratoconus were interpreted as postkeratoplasty, and one postkeratoplasty map was interpreted as advanced keratoconus. One astigmatism map was diagnosed as a map of a normal cornea. Five maps were classified as unknown (two normal corneas, one mild and one advanced keratoconus, and one keratoplasty).

DISCUSSION

Automated interpretation of topographic maps would be helpful to inexperienced observers for improving their decision making performance in the differential diagnosis of topographic patterns to a level approaching that of more experienced corneal topography users. Also, for experienced examiners, machine interpretation would be useful to provide an objective correlate of clinical diag-

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FIGURE 3. (*top left*) Color-coded map of a normal cornea. Its output strength by the neural network is as follows:

Category	NRM	AST	KC1	KC2	KC3	PRK	KP
Output strength	0.99	0	0	0	0	0	0

Note that the neural network demonstrates an appropriate combination of outputs in this case.

FIGURE 4. (*bottom left*) Color-coded map of moderate keratoconus. Its output strength by the neural network is as follows:

Category	NRM	AST	KC1	KC2	KC3	PRK	KP
Output strength	0	0	0.03	0.99	0	0	0

As in Figure 3, the neural network shows a correct combination of output strengths.

FIGURE 5. (*top right*) Color-coded map of a postkeratoplasty eye. Its output strength by the neural network is as follows:

Category	NRM	AST	KC1	KC2	KC3	PRK	KP
Output strength	0	0	0	0	0.55	0	0

In this case, the neural network has misclassified the map as advanced keratoconus. Note, however, that the map shows localized steepening and asymmetry.

FIGURE 6. (*bottom right*) Color-coded map of mild keratoconus. Its output strength by the neural network is as follows:

Category	NRM	AST	KC1	KC2	KC3	PRK	KP
Output strength	0	0.99	0.06	0.08	0	0	0

Although this map shows abnormal central steepening and a lazy 8 figure, it is misinterpreted as astigmatism.

nosis and to provide quantitative information that may be used for comparison among patients or for monitoring serial changes in the progression of disease or over the course of treatment. Although interpretation by experienced examiners could be influenced by various conditions, such as emotions, concentration, or memories, a computer-assisted decision support system always provides objective information.

Objective classification or quantification of the topographic pattern has been described with the photokeratoscope and the videokeratoscope.[16,18–20] However, the subjects in these studies were limited to a few categories, such as normal cornea, regular astigmatism, and keratoconus. To date, no method has been described that is capable of accurate classification of a wide variety of topographic patterns. In contrast, a human expert can easily classify various topographic patterns by visual inspection of the topographic maps. In human interpretation, the cognitive processes involved in the complicated pattern analysis of color-coded maps are developed through clinical training and experience. A large set of topographic images (input) and their correct answers (output) are used to learn the relationships between input and output data. Neural networks can "learn" in an analogous manner. Neural networks also compare the output to the correct answer and change weights between neurons repetitively to produce better approximations. Relationships between certain input patterns and corresponding outputs modulated by the hidden layer(s) are memorized as a matrix of weights.

The backpropagation method, which is used in this study, is a supervised, multilayer, feed-forward network.[17] It is one of the most commonly used neural networks because the training process can be achieved relatively easily and the interpretation of the results is also plain. Once a network is trained with the backpropagation method, it always produces the same outputs in response to a set of given inputs. This ability to provide reproducible answers for the same subjects seems to be desirable for a decision support system in medicine.

Although computer-assisted diagnosis can be accomplished with other methods, such as multivariate analysis (discriminant analysis) or a conventional artificial intelligence approach (expert system), neural

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networks offer advantages over these methods.[8,10] Neural networks show a unique ability to detect features that are hidden in the input data but are not explicitly formulated as input.[8] Neither multivariate analysis nor expert systems can use hidden relationships in the input data. Also, the learning process of the neural network is based solely on the data supplied; no information is required about relationships, rules, or other logic structures between input and output data, whereas the expert system involves an extensive set of rules, deductive decision making, and step-by-step logical operation.

For neural networks, establishment of input data is one of the key factors in developing a successful interpretation system. In this study, a neural network was developed to classify topographic maps into diagnostic categories based on patterns seen in topographic indices. In place of raw topographic data, quantitative indices were used to describe topographic features to reduce the number of parameters. Topographic indices that can extract the original information with fewer parameters are thought to be effective in simplifying the architecture and the training process of neural networks. In our study, 11 indices represent different characteristics seen in topographic maps. The DSI and OSI are sensitive to a localized steepening in the periphery, and the CSI is sensitive to aspheric abnormalities. SAI represents topographic asymmetry, and SRI, IAI, and AA describe the irregularity of corneal power from different aspects. Also, SimK1, SimK2, and MinK show corneal steepening and the amount of regular and irregular astigmatism, respectively. Each topographic category has a unique distribution of indices, as seen in Table 3. For example, the values of SimK1, SimK2, and MinK in normal corneas resemble each other, and all the other indices, except AA, have low values. Compared with the normal category, the astigmatism category has a large difference between SimK1 and SimK2, and both DSI and SDP are larger than they are for the normal category. Keratoconus shows higher SAI, SRI, and OSI values than astigmatism.

As shown in the results, the neural network was successfully trained to classify all topographic maps in the training set and classified 80% of the maps correctly in the independent test set. This result is fairly good for a preliminary demonstration of the classification of seven categories of topographic abnormalities. Neural networks used for cancer diagnosis[9] or glaucoma interpretation[11] have shown similar or less accuracy (80% and 65%, respectively), although their outputs are binary. Misinterpretations were found between mild keratoconus and with-the-rule astigmatism, between with-the-rule astigmatism and normal cornea, between advanced keratoconus and keratoplasty, and between various levels of severity in keratoconus. Unacceptable mistakes, such as interpreting normal as advanced keratoconus, did not occur. Although there were five maps in which no output neuron had a value > 0.5 (classified as unknown pattern), the output strength for the correct category showed the highest values in all cases. It may be possible to control sensitivity and specificity according to the purpose of the neural network by changing the cut-off value.

Specificity was consistently high, but sensitivity showed a wide variation. The reason may be related to the smaller numbers in the sensitivity ratios compared to the ratios for specificity and accuracy. The two categories with the lowest sensitivity also had the smallest numbers of any of the calculations.

The difference in accuracy between the training set (100%) and the test set (80%) was statistically significant (chi-square analysis; $P = 0.0001$). The reduction in accuracy between the two sets suggests that the neural network is not sufficiently generalized. One possible explanation for this disparity is that the neural network may be categorizing by memorizing some unimportant features in the maps during the training process. Further optimization might be obtained by modifying the input parameters, the architecture of the neural network, or both.

Because the ability of a neural network depends on its "training" or "experience" for accurate response, the results will also degenerate significantly when there are insufficient data in a given category. Perhaps the number of maps chosen for the training set was too limited, and this limitation may have been the reason for some of the erroneous responses obtained for the test set (Table 5).

Another source of error may lie in the difference in feature extraction between visual inspection of the color-coded map and this neural network. Although visual inspection reveals various patterns from the observation of the entire corneal image, this neural network accesses topographic features with 11 empirical indices. This implies that the network has a set of input data that is different from (and possibly more limited than) the data acquired by human inspection.

To improve generalization and accuracy, sufficient varieties and numbers of maps in each category must be available, and new indices may be necessary. The responses of the current network to maps of disorders other than those in the seven "training" categories or to atypical maps are unknown, because the network was developed only for maps that have a typical topographic appearance for one of these particular corneal abnormalities.

Despite the many questions awaiting resolution, these preliminary results suggest that the use of neural networks for the interpretation of corneal topography deserves further development. This type of system of-

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fers significant potential advantages as an integral part of the interpretation of corneal topography for the diagnosis of anomalies of corneal shape.

Key Words

neural network, artificial intelligence, corneal topography, image analysis, refractive surgery

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